

6 (**long**) Figure 5 shows a gas turbine power plant. Air enters the compressor at State 1 ($p_1 = 1$ bar, $T_1 = 300$ K) with a mass flow rate of 10 kg s^{-1} . It is compressed to State 2 ($p_2 = 10$ bar), where the flow is divided into two streams. The main stream passes through a heater to State 3 ($p_3 = 10$ bar, $T_3 = 1400$ K) and then through a turbine to State 4 ($p_4 = 1.2$ bar). A bypass stream passes through a throttle valve to State 5 ($p_5 = 1.2$ bar). The two streams are mixed adiabatically to produce a single flow at State 6 ($p_6 = 1$ bar). The turbine and compressor are both adiabatic and reversible, and the net power output from the plant is $\dot{W}_{\text{net}} = 3$ MW. Changes in kinetic and potential energy between the numbered states may be neglected and air may be treated as a perfect gas. The plant operates steadily.

(a) Calculate the compressor exit temperature T_2 and the power input to the compressor. [5]

(b) Calculate the turbine exit temperature T_4 , the mass flow rate through the turbine, and the efficiency of the power plant. [10]

(c) Calculate the mixer exit temperature T_6 and the rate of entropy generation in the mixer. [10]

(d) Sketch a T - s diagram indicating all processes undergone by the streams (*i.e.*, states 1–6). Use solid lines to denote paths through equilibrium states and dotted lines where there are no paths through equilibrium states. [5]

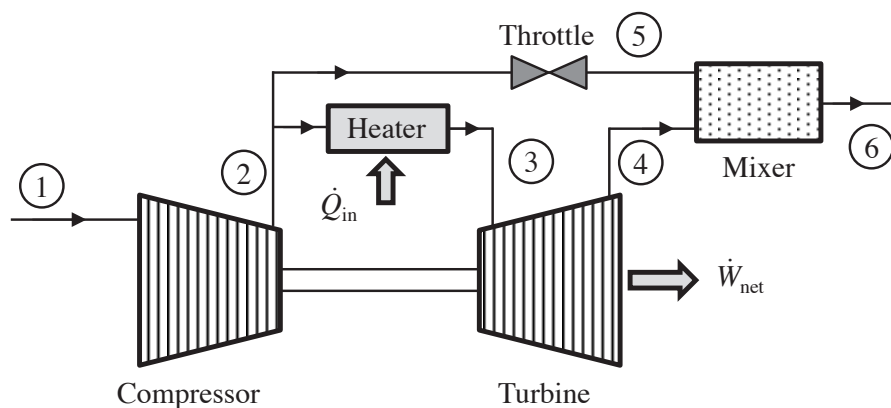


Fig. 5